



U.S. Department
of Transportation

Research and
Special Programs
Administration

FEB 14 1996

400 Seventh Street, S.W.
Washington, D.C. 20590

DOT-E 9357 (EXTENSION)
ORIGINAL June 6, 1985
CORRECTED COPY

In accordance with 49 CFR 107.105 of the Department of Transportation (DOT) Hazardous Materials Regulations DOT-E 9357 is hereby extended for the party(ies) listed below by changing the expiration date in paragraph 10 to January 31, 1998. This change is effective from the issue date of this extension. All other terms of the exemption remain unchanged.

This extension applies only to party(ies) listed below based on the application(s) received in accordance with 49 CFR 107.105. This extension constitutes a necessary part of this exemption and must be attached to it.

for Marilyn E. Manis
Alan I. Roberts
Associate Administrator
for Hazardous Materials Safety

FEB 14 1996

(DATE)

Dist: FHWA FRA USCG

EXEMPTION HOLDER

APPLICATION DATE

Transamerica Leasing, Inc.
Purchase, NY

February 7, 1996



U.S. Department
of Transportation

Research and
Special Programs
Administration

400 Seventh St., S.W.
Washington, D.C. 20590

DOT-E 9357
(CORRECTED COPY)

1. Dynatrans Aktiebolag Sweden, Gothenburg, Sweden (U.S. Agent: Sea Container Americas Inc., New York, N.Y.), is hereby granted an exemption from those provisions of this Department's Hazardous Materials Regulations specified in paragraph 5 below to offer the packaging described in paragraph 7 below for use in the transportation of the liquefied compressed gases described in paragraph 3 below in commerce subject to the requirements specified herein. This exemption authorizes the use of a non-DOT specification IMO Type 5 portable tank, and provides no relief from any regulations other than as specifically stated.

2. BASIS. This exemption is based on Dynatrans Aktiebolag Sweden's application dated November 15, 1984, and additional information dated March 8, 1985, submitted in accordance with 49 CFR 107.103 and the public proceeding thereon.

3. HAZARDOUS MATERIALS (Descriptor and class). Difluoroethane classed as a flammable gas, dichlorodifluoromethane classed as a non-flammable gas; other liquefied compressed gases specifically identified to; and acknowledged in writing by, the Office of Hazardous Materials Regulation (OHMR) prior to the first shipment.

4. PROPER SHIPPING NAME (49 CFR 172.101). The chemical name, or generic description, as appropriate.

5. REGULATION AFFECTED. 49 CFR 173.315, 178.245.

6. MODES OF TRANSPORTATION AUTHORIZED. Motor vehicle, rail freight, cargo vessel.

7. SAFETY CONTROL MEASURES.

- a. Packaging prescribed is a non-DOT specification portable tank, mounted in ISO, frame, designed and constructed in accordance with Dynatrans Aktiebolag drawings 0-0015, Issue 2, 0-0013, Issue 1, 0-0014, Issue 1, other drawings, technical specifications and calculations on file with the OHMR, and in compliance with the following:

(1) Code - Complies with DOT Specification 51 except that tanks are not ASME Code "U" stamped and have bottom outlets; IMO Type 5.

(2) Insulation - None, equipped with a sun shield.

(3) Water capacity (U.S. Gallons) - 4,544

(4) Material - Swedish steel OX 540 E equivalent to ASTM SA 537 Class 1; Yield strength - 56,660 psig; Tensile strength - 76,050

	(outside dia.)	X (length)	X (thickness)
(5) Tank Size (inches)	78.74	236.22	0.443(min.)

Head Thickness - 0.4724 (min.)
Weld Joint Efficiency - 1.0
Corrosion Allowance - 0.0394

- (6) Design Pressure (PSIG) - 193
Note: Design pressure means "maximum allowable working pressure (MAWP)" as used in the ASME Code.
- (7) Test Pressure, Minimum (PSIG) - 289.5
- (8) Openings - One(1) - 24.4 inch diameter opening for the manhole which contains the pressure relief devices on the top; one(1) - 12.6 inch diameter opening for the rotary gauge on the head; one(1) - 8.3 inch diameter opening for the liquid phase valve and one(1) - 6.5 inch diameter opening for the vapor phase valve on the bottom. NOTE: Each bottom outlet valve must be provided with a shear section that meets the requirements of 49 CFR 178.337-12.
- (9) Tank surface area (square feet) - 441
- (10) Pressure Relief Devices - Two(2) - 2 1/2 inch diameter spring loaded pressure relief valves set at 202 psig. Total relief device capacity is 952,800 SCFH.
- (11) G-Loadings: Vertical down 4 : Vertical up 2 :
Longitudinal 4 : and Transverse 2 .
- (12) Maximum Gross Weight (pounds) - 67,200.
- (13) Maximum Commodity Weight (pounds) - 54,678.
- (14) Tare Weight (pounds) - 12,522.
- (15) Design Specific Gravity - 1.44
- (16) Design Temperature Range ($^{\circ}$ F) - -40 to 140.

8. SPECIAL PROVISIONS.

- a. Shippers may use the packaging covered by this exemption pursuant to the 49 CFR 173.22a.
- b. A copy of this exemption must be carried aboard each vessel used to transport packages covered by this exemption.
- c. Each portable tank must be plainly marked on both sides near the middle, in letters at least two inches high on a contrasting background, "DOT-E 9357".
- d. Hydrostatic test certificates for each tank must be maintained by the owner or manufacturer at its principal business office and be made available to any representative of the DOT upon request.

- e. No product may be shipped that has venting requirements exceeding 952,800 SCFH, or has a specific gravity greater than 1.80. The venting capacity required for each product must be determined by the flow formulas contained in Compressed Gas Association (CGA) pamphlet 5-1.2.
- f. The tank must be filled by weight in accordance with the provisions of 49 CFR 173.315.
- g. Each tank must be (i) visually inspected prior to each trip to insure that it has not been damaged on the previous trip; and (ii) retested and reinspected once every five years in accordance with 49 CFR 173.32 as prescribed for DOT Specification 51 portable tanks.
- h. Portable tanks may not be transported in container-on-flat car (COFC) or trailer-on-flat car (TOFC) service except under conditions approved by the Associate Administrator for Safety, Federal Railroad Administration.
- i. DOT-E 9357 must be stamped on the metal manufacturer's data plate on the line which reads "U.S. DOT Specification No.".
- j. For each portable tank, the manufacturer must prepare a certificate which must be signed by a responsible official of the manufacturer and a National Board of Boiler and Pressure Vessel Inspectors Authorized Inspector certifying that the portable tank is designed and constructed in accordance with the ASME Code and this exemption. The certificate for the first portable tank fabricated must be submitted to the OHMR prior to the initial shipment.

9. REPORTING REQUIREMENTS.

- a. The owner of the tanks covered by this exemption shall provide to the OHMR a list of all shippers who have shipped under this exemption and a shipping experience report every six months from the effective date of this exemption. The shipping experience report shall contain approximate number of shipments made during the previous six months and any difficulties encountered.
- b. The holder or parties to this exemption, as identified in paragraph 1 above, shall contact the OHMR immediately after any of the tanks covered by this exemption are sold to another party.
- c. Shippers who ship under the terms of this exemption shall report any incident involving loss of contents of the tanks described herein to the OHMR as soon as practicable.

10. EXPIRATION DATE. March 31, 1987.

Issued at Washington, D.C.



Alan I. Roberts
Associate Director for
Hazardous Materials Regulation
Materials Transportation Bureau

JUN 6 1985

(DATE)

Address all inquiries to: Associate Director for Hazardous Materials Regulation, Materials Transportation Bureau, Research and Special Programs Administration, U.S. Department of Transportation, Washington, D.C. 20590. Attention: Exemptions Branch.

Dist: USCG, FHWA, FRA.



U.S. Department
of Transportation

Research and
Special Programs
Administration

FEB 14 1996

400 Seventh Street, S.W.
Washington, D.C. 20590

DOT-E 9357 (EXTENSION)
ORIGINAL June 6, 1985
CORRECTED COPY

In accordance with 49 CFR 107.105 of the Department of Transportation (DOT) Hazardous Materials Regulations DOT-E 9357 is hereby extended for the party(ies) listed below by changing the expiration date in paragraph 10 to January 31, 1998. This change is effective from the issue date of this extension. All other terms of the exemption remain unchanged.

This extension applies only to party(ies) listed below based on the application(s) received in accordance with 49 CFR 107.105. This extension constitutes a necessary part of this exemption and must be attached to it.

for Marilyn E. Morris
Alan I. Roberts
Associate Administrator
for Hazardous Materials Safety

FEB 14 1996

(DATE)

Dist: FHWA FRA USCG

EXEMPTION HOLDER

APPLICATION DATE

Transamerica Leasing, Inc.
Purchase, NY

February 7, 1996



U.S. Department
of Transportation

Research and
Special Programs
Administration

400 Seventh St., S.W.
Washington, D.C. 20590

DOT-E 9357
(CORRECTED COPY)

1. Dynatrans Aktiebolag Sweden, Gothenburg, Sweden (U.S. Agent: Sea Container Americas Inc., New York, N.Y.), is hereby granted an exemption from those provisions of this Department's Hazardous Materials Regulations specified in paragraph 5 below to offer the packaging described in paragraph 7 below for use in the transportation of the liquefied compressed gases described in paragraph 3 below in commerce subject to the requirements specified herein. This exemption authorizes the use of a non-DOT specification IMO Type 5 portable tank, and provides no relief from any regulations other than as specifically stated.

2. BASIS. This exemption is based on Dynatrans Aktiebolag Sweden's application dated November 15, 1984, and additional information dated March 8, 1985, submitted in accordance with 49 CFR 107.103 and the public proceeding thereon.

3. HAZARDOUS MATERIALS (Descriptor and class). Difluoroethane classed as a flammable gas, dichlorodifluoromethane classed as a non-flammable gas; other liquefied compressed gases specifically identified to; and acknowledged in writing by, the Office of Hazardous Materials Regulation (OHMR) prior to the first shipment.

4. PROPER SHIPPING NAME (49 CFR 172.101). The chemical name, or generic description, as appropriate.

5. REGULATION AFFECTED. 49 CFR 173.315, 178.245.

6. MODES OF TRANSPORTATION AUTHORIZED. Motor vehicle, rail freight, cargo vessel.

7. SAFETY CONTROL MEASURES.

- a. Packaging prescribed is a non-DOT specification portable tank, mounted in ISO, frame, designed and constructed in accordance with Dynatrans Aktiebolag drawings 0-0015, Issue 2, 0-0013, Issue 1, 0-0014, Issue 1, other drawings, technical specifications and calculations on file with the OHMR, and in compliance with the following:

(1) Code - Complies with DOT Specification 51 except that tanks are not ASME Code "U" stamped and have bottom outlets; IMO Type 5.

(2) Insulation - None, equipped with a sun shield.

(3) Water capacity (U.S. Gallons) - 4,544

(4) Material - Swedish steel OX 540 E equivalent to ASTM SA 537 Class 1; Yield strength - 56,660 psig; Tensile strength - 76,050

	(outside dia.) X (length) X (thickness)		
(5) Tank Size (inches)	78.74	236.22	0.443(min.)

Head Thickness - 0.4724 (min.)

Weld Joint Efficiency - 1.0

Corrosion Allowance - 0.0394

- (6) Design Pressure (PSIG) - 193

Note: Design pressure means "maximum allowable working pressure (MAWP)" as used in the ASME Code.

- (7) Test Pressure, Minimum (PSIG) - 289.5

- (8) Openings - One(1) - 24.4 inch diameter opening for the manhole which contains the pressure relief devices on the top; one(1) - 12.6 inch diameter opening for the rotary gauge on the head; one(1) - 8.3 inch diameter opening for the liquid phase valve and one(1) - 6.5 inch diameter opening for the vapor phase valve on the bottom. NOTE: Each bottom outlet valve must be provided with a shear section that meets the requirements of 49 CFR 178.337-12.

- (9) Tank surface area (square feet) - 441

- (10) Pressure Relief Devices - Two(2) - 2 1/2 inch diameter spring loaded pressure relief valves set at 202 psig. Total relief device capacity is 952,800 SCFH.

- (11) G-Loadings: Vertical down 4 : Vertical up 2 :
Longitudinal 4 : and Transverse 2 .

- (12) Maximum Gross Weight (pounds) - 67,200.

- (13) Maximum Commodity Weight (pounds) - 54,678.

- (14) Tare Weight (pounds) - 12,522.

- (15) Design Specific Gravity - 1.44

- (16) Design Temperature Range ($^{\circ}$ F) - -40 to 140.

8. SPECIAL PROVISIONS.

a. Shippers may use the packaging covered by this exemption pursuant to the 49 CFR 173.22a.

b. A copy of this exemption must be carried aboard each vessel used to transport packages covered by this exemption.

c. Each portable tank must be plainly marked on both sides near the middle, in letters at least two inches high on a contrasting background, "DOT-E 9357".

d. Hydrostatic test certificates for each tank must be maintained by the owner or manufacturer at its principal business office and be made available to any representative of the DOT upon request.

e. No product may be shipped that has venting requirements exceeding 952,800 SCFH, or has a specific gravity greater than 1.80. The venting capacity required for each product must be determined by the flow formulas contained in Compressed Gas Association (CGA) pamphlet S-1.2.

f. The tank must be filled by weight in accordance with the provisions of 49 CFR 173.315.

g. Each tank must be (i) visually inspected prior to each trip to insure that it has not been damaged on the previous trip; and (ii) retested and reinspected once every five years in accordance with 49 CFR 173.32 as prescribed for DOT Specification 51 portable tanks.

h. Portable tanks may not be transported in container-on-flat car (COFC) or trailer-on-flat car (TOFC) service except under conditions approved by the Associate Administrator for Safety, Federal Railroad Administration.

i. DOT-E 9357 must be stamped on the metal manufacturer's data plate on the line which reads "U.S. DOT Specification No.".

j. For each portable tank, the manufacturer must prepare a certificate which must be signed by a responsible official of the manufacturer and a National Board of Boiler and Pressure Vessel Inspectors Authorized Inspector certifying that the portable tank is designed and constructed in accordance with the ASME Code and this exemption. The certificate for the first portable tank fabricated must be submitted to the OHMR prior to the initial shipment.

9. REPORTING REQUIREMENTS.

a. The owner of the tanks covered by this exemption shall provide to the OHMR a list of all shippers who have shipped under this exemption and a shipping experience report every six months from the effective date of this exemption. The shipping experience report shall contain approximate number of shipments made during the previous six months and any difficulties encountered.

b. The holder or parties to this exemption, as identified in paragraph 1 above, shall contact the OHMR immediately after any of the tanks covered by this exemption are sold to another party.

c. Shippers who ship under the terms of this exemption shall report any incident involving loss of contents of the tanks described herein to the OHMR as soon as practicable.

10. EXPIRATION DATE. March 31, 1987.

Issued at Washington, D.C.



Alan I. Roberts
Associate Director for
Hazardous Materials Regulation
Materials Transportation Bureau

JUN 6 1985

(DATE)

Address all inquiries to: Associate Director for Hazardous Materials
Regulation, Materials Transportation Bureau, Research and Special Programs
Administration, U.S. Department of Transportation, Washington, D.C. 20590.
Attention: Exemptions Branch.

Dist: USCG, FHWA, FRA.